

Quantifying Information Extracted from Gravitational-Wave Observations

AUTHOR

Ömer Faruk Soydan 2555472

ADVISOR

Assist. Prof. Dr. Doğa Veske

FIELD

Computational Data Analysis · Theoretical Testing

01 Introduction

Every detection sharpens our knowledge of a source. We measure that gain in **bits** — the **KL divergence** from posterior to prior — then ask **how much** each detection extracts, **how** it is distributed, and whether a **compact law** beats the classic **Flanagan–Hughes** estimate.

EXACT INFORMATION · KL DIVERGENCE

$$I_{\text{source}} = \int d\theta \, p(\theta|s) \log^2 \frac{p(\theta|s)}{p(\theta|p(s))}$$

The reduction in uncertainty about the 15 source parameters, computed directly from public posterior samples — the central quantity of this project.

FLANAGAN–HUGHES HIGH-SNR ESTIMATE

$$I_{\text{source}} \approx \frac{N_{\text{param}}}{2} \log^2 \left(1 + \frac{\rho^2}{N_{\text{param}}} \right)$$

Gaussian-noise, linear-waveform limit. $N_{\text{param}} = 15 \cdot \rho = \text{matched-filter SNR}$. Called "a rather crude approximation" by its own authors.

41.5 bit Flanagan–Hughes prediction for a **GW150914**-like system — our reference point to beat.

02 Methods

Direct density estimation in 15-D fails — the **curse of dimensionality**. Every standard estimator breaks down for one shared reason:

ESTIMATOR	TYPE	RESULT & STATUS
k-NN	Nearest-neighbour	25.9 → 104.9 bit never converges
Multivariate Gaussian	Parametric	≈ 41.2 bit cond. # ~10 ⁹
KDE	Smooth kernel	— infeasible in 15-D
np.histogramdd	Binning	5 ¹⁵ ≈ 3×10 ¹⁰ cells ~240 GB RAM

OUR FIX → SPLIT INTO FIVE ≤5-D BLOCKS

G1 Masses	G2 Spin mag. & tilts	G3 Azimuthal spins	G4 Distance & orient.	G5 Sky · time · phase
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Within each block, **three independent estimators** (KDE-Scott, KDE-Silverman, k=1 NN) are cross-checked, and a **hybrid prior** removes the stored-prior support artefact in distance and mass.

TOTAL-CORRELATION CORRECTION RESTORES BETWEEN-BLOCK COUPLING

$$I_{\text{joint}} \approx \sum_g I_{\text{group}}(g) + (\text{TC}_{\text{post}} - \text{TC}_{\text{prior}})$$

03 Results

39 bit

Measured for GW150914 — about **2 bit below** the 41.5-bit prediction.

6.7–44.6 bit

Range across all **35 events**, quietest to loudest.

∝ ln(SNR)

Information grows as the **log of SNR** — exactly the FH high-SNR limit.

0.88 R²

Three-term law (SNR, mass, detectors) explains **88%** of variance.

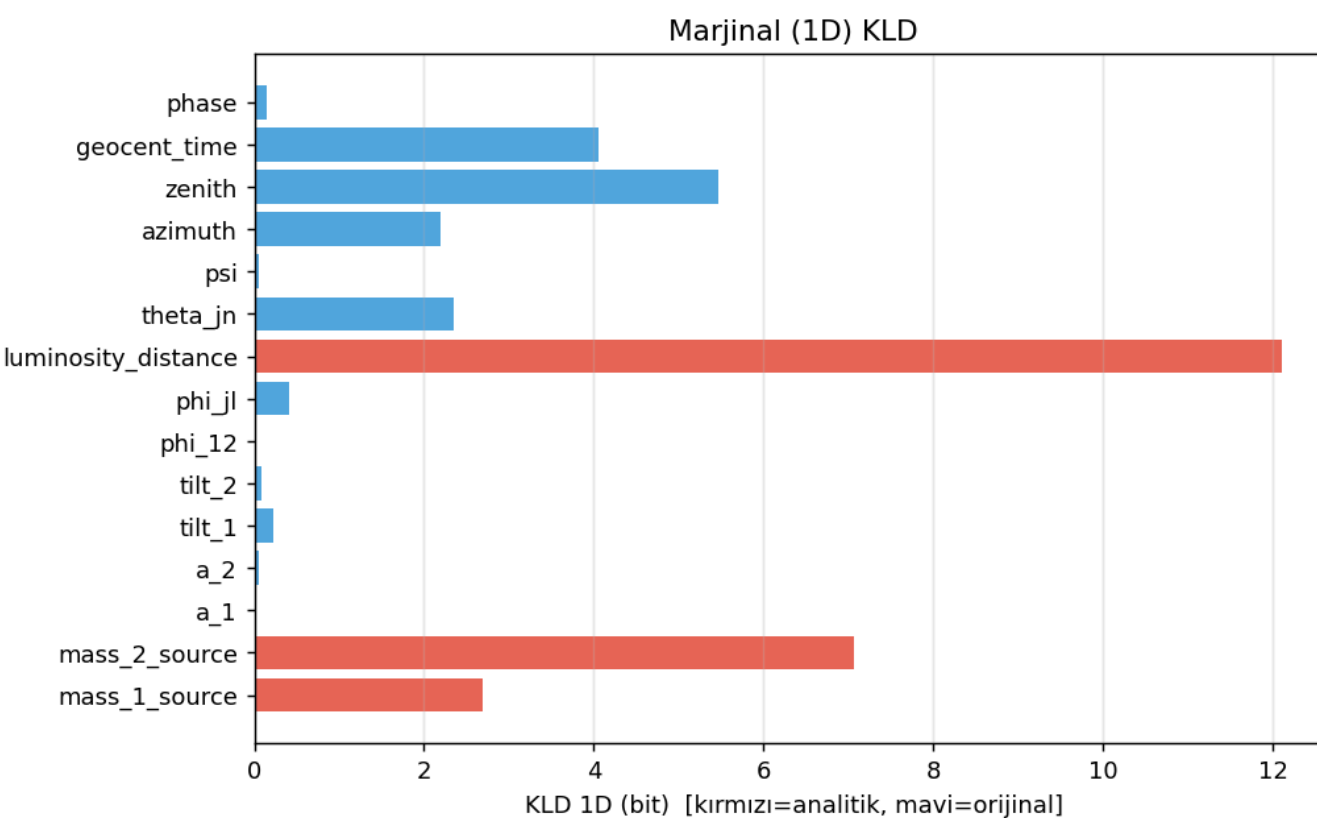
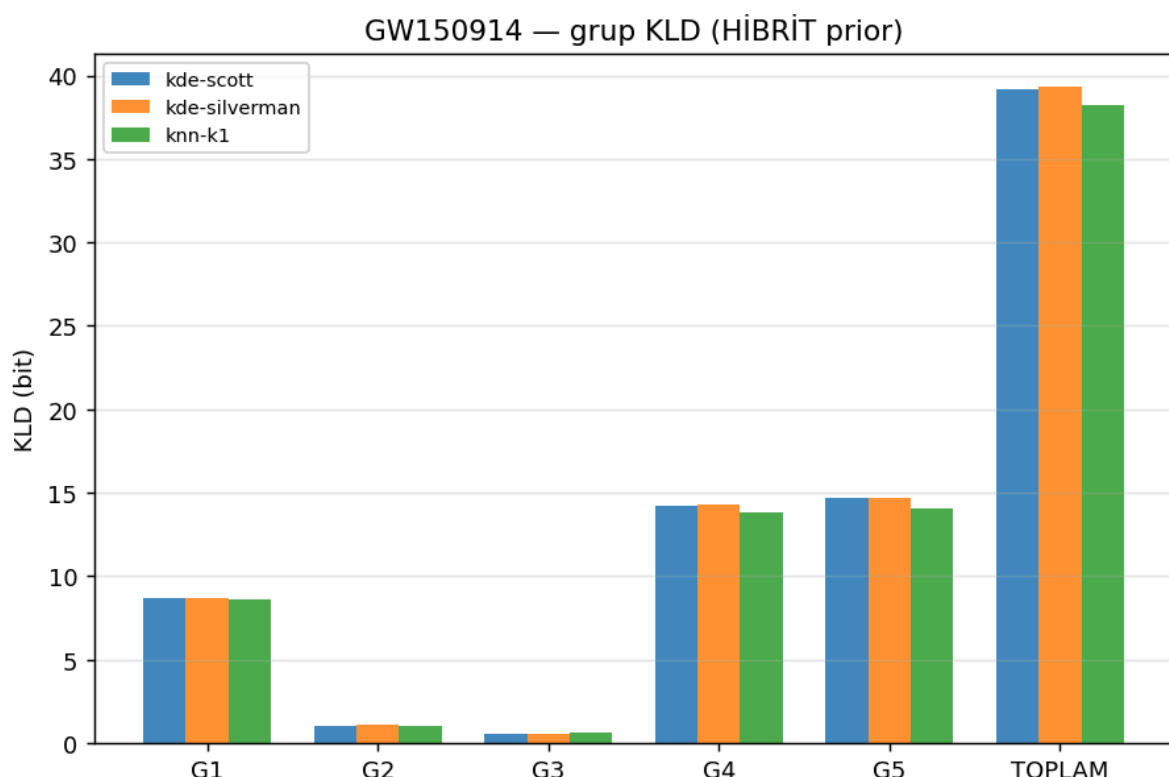


FIG 1 · Single event, GW150914 — ≈ 39 bit. Per-block (left) and marginal (right) KL. The three estimators — KDE-Scott, KDE-Silverman, k=1 NN — overlap in every block → a method-independent total. Luminosity distance and the masses dominate the information.

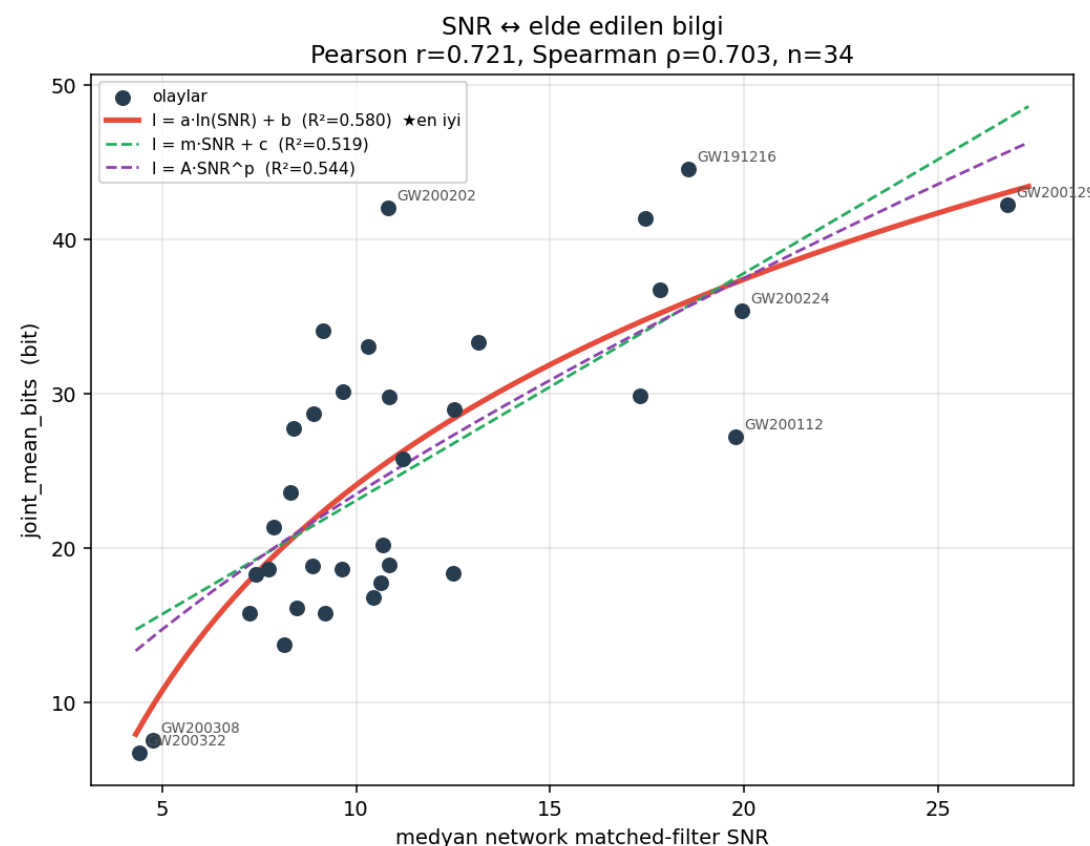


FIG 2 · Population scaling, 35 events. Joint information vs. network SNR. A logarithmic law $I = a \cdot \ln(\text{SNR}) + b$ ($r = 0.72$) beats the linear & power-law fits.

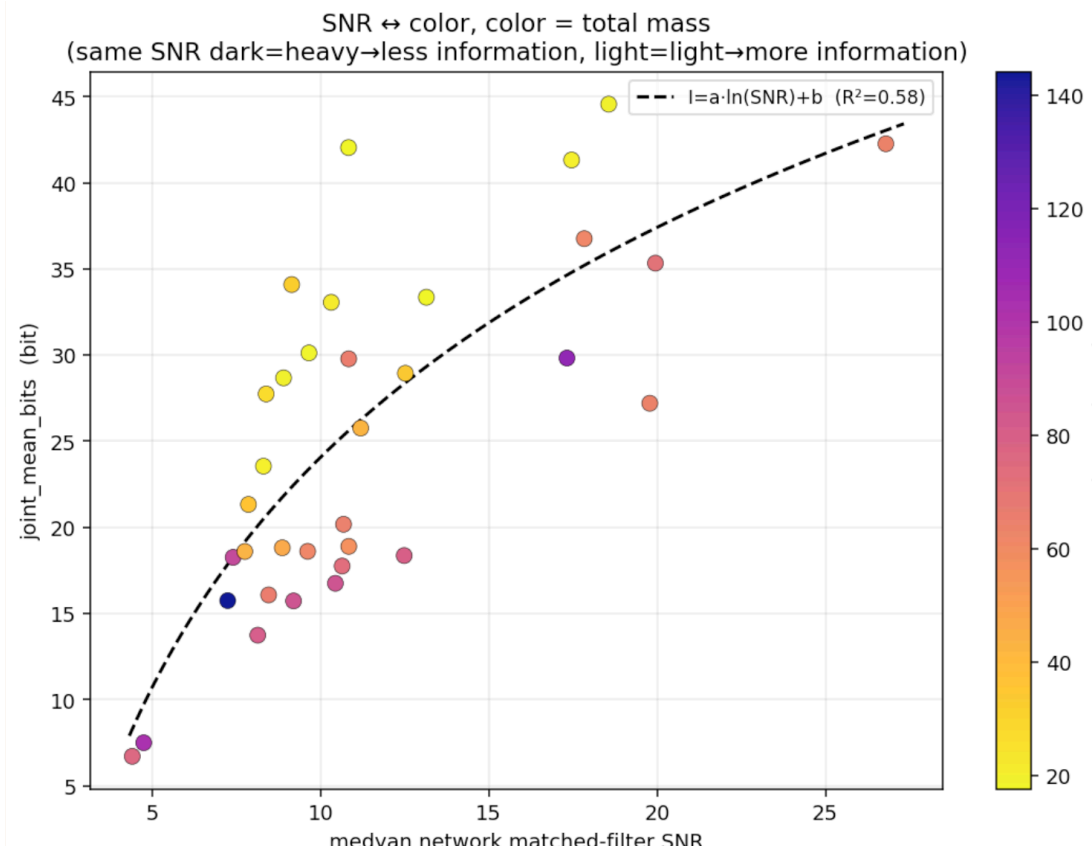


FIG 3 · Lighter binaries carry more. Colour = total mass. At fixed SNR, lighter binaries (yellow) carry more information than heavy ones (dark) — the second driver behind SNR.

04 · CONCLUSION

Flanagan–Hughes is accurate, but slightly optimistic.

It reproduces the right order of magnitude and the right functional scaling, while overstating the absolute value by ~2 bit — traceable to the non-Gaussianity of real posteriors. The most informative events are **louder, lighter binaries seen by more detectors**.

BEST EMPIRICAL LAW

$$I = a \ln(\text{SNR}) + c \ln(M) + d n_{\text{det}}$$

Explained variance **R² = 0.88** across the catalogue.